Consolidated Probationary Period Report

ASTANA IT UNIVERSITY

School of Software Engineering · Software Engineering programme
Group SE-2430

Smart Finance

Development of an Intelligent Application for Personal Finance Planning and Expense Analysis

Submission	Identification
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Stage / report date	Probationary period 1 · 24 October 2026
Frozen source	Tag: **prob-period-1** 8dd1d51679fe4672bb00e198b3a55ccea2d0c774
Code / interface checks	50 automated tests: OK; synthetic analytics reproduced; emulator UI demonstration confirmed by the team

https://github.com/KaberDaryn/smart-finance-project/tree/prob-period-1

https://github.com/users/KaberDaryn/projects/1

Submission boundary. Code findings refer to the verified tag above. Supplied UI/board screenshots are supporting snapshots, not proof of a matching build or completed acceptance. This report and its newly prepared academic documents/diagram sources were not present in the inspected tag. Repository publication and review remain open.

Purpose: consolidate the 11 required sections, including the supplied teacher review and five user-feedback summaries. Source-verified checks, reported qualitative feedback and planned evaluation are distinguished.

Editable diagram sources, the support package, verification output and provenance are embedded as PDF file attachments. Use a PDF reader that supports attachments; see the final page.

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Reading the evidence

Label	Meaning
Verified at tag	Checked against source 8dd1d51679fe or reproduced locally on that exact source.
Supporting snapshot	Supplied image or live task record; date/build may differ from the frozen source.
Planned / proposed	Acceptance target or study protocol; no completed result or supervisor approval is implied.
Not established	Required evidence or accessible publication was not available for this submission.

Source citations [R1–R16], [P1–P5] and named repository links are given alongside the relevant claims. All public links are written in full and are clickable.

1. Project proposal

Problem statement and intended users

Personal finance records may be fragmented across cash receipts and manual notes. Repeated entry can discourage consistent recording, while automatic extraction may introduce incorrect values. The engineering problem is to combine capture, validation, user review, expense reflection and planning without allowing uncertain AI output to commit financial records automatically. Five supplied user-feedback summaries support these concerns and identify adoption barriers (§2); they do not quantify prevalence or product impact [R1, R9].

Intended users are students and individuals recording KZT income/cash expenses, reviewing category spending, setting monthly limits and tracking manually entered savings progress. No bank connection, payment execution or personalised investment advice is proposed.

Aim and objectives

Aim: develop and verify a reproducible local personal-finance prototype with account-scoped records, planning tools and editable analytical assistance.

Objective	Expected outcome / evidence
1 · Define and trace scope	Prioritised FR/NFR, acceptance criteria and named verification cases (§6, §9).
2 · Integrate components	Client → authenticated JSON API → relational storage; reproducible setup (§7–8).
3 · Protect financial correctness	Integer minor units, valid dates, account isolation and atomic import/save operations.
4 · Add optional assistance	Consented receipt proposals, editable category suggestions and manual fallback; no auto-save.
5 · Evaluate distinct outcomes	Functional tests separately from field accuracy, classifier macro-F1, forecast error and human task time.

Scope and engineering contribution

In scope: accounts; dated KZT income/expense records; history/search; category/month summaries; CSV import; budgets/goals; optional Gemini receipt assistance; per-user confirmed merchant/category memory; English-description Naive Bayes and a three-completed-month mean forecast. Integration, exact data handling, failure recovery and traceability provide engineering complexity; no new learning algorithm is claimed.

Excluded: bank sync, real transfers, exchange-rate service, autonomous advice, independent QR decoding, guaranteed cloud uptime, production certification and validated multilingual classification. Foreign receipts require a user-entered KZT equivalent. Current status: working local code and tests, with team-confirmed emulator interface checks; qualitative user feedback is now recorded (§2), while full acceptance and measured research-quality evidence remain to be completed.

2. Evidence of relevance

Date: 24 October 2026. Method: qualitative online survey through social networks, as reported by the team. Five responses represent different recording habits and expectations. U1–U5 identify the anonymous response summaries. The table translates the supplied Russian comments and retains both positive and negative findings. No timed usability or representative population study is claimed.

User perspective	What was valued	Concern / adoption barrier
U1 · Student who rarely tracks spending	Simple sign-in, a short expense form and easy history search.	Opening the app manually for every expense may become a burden. Without reminders or quick entry, the user expects to stop using it.
U2 · Cash payer who keeps receipts	Photograph → review → save is appealing. Explicit confirmation before recording increases trust.	Uncertain performance on crumpled receipts and Kazakhstan retailers; repeated errors may make manual entry easier. External image processing raises privacy concerns.
U3 · Person saving towards a goal	Values budgets, category limits, overspending warnings and percentage-based savings progress.	Saved amounts need manual updates and are not tied to a real account. Wants an indication of how long available money will last, not only a goal amount.
U4 · Existing banking-app user	Values exact KZT minor-unit handling, CSV import and separation between accounts.	Banking apps already contain and categorise this user’s expenses. Duplicate entry and the absence of bank sync are the main barriers.
U5 · Analytics-focused user	Likes category/month summaries, a next-month forecast and editable category suggestions.	A three-month average is considered too simple for irregular spending. Inaccurate suggestions reduce trust; the user wants an explanation for each suggestion.

Interpretation of the feedback

The strongest supported direction is **reviewable receipt-assisted entry for people who retain cash receipts**, with a low-friction manual fallback. U1 and U4 highlight different adoption barriers: habitual effort and duplicate bookkeeping. U3 and U5 request stronger planning/analytical support. These perspectives do not demonstrate universal demand or justify expanding the MVP into banking integration.

Evidence record [U1–U5]: five qualitative social-media survey responses collected on 24 October 2026, according to the team. This is a small, non-representative sample; no percentages or population-wide conclusions are drawn. The social platform, recruitment/consent record and original survey messages were not supplied. Full provided comments are preserved in USER_FEEDBACK.md inside the support package. The survey is not relabelled as five in-person interviews.

2.1 Findings translated into requirements and decisions

Finding / source	Response and traceability	Owner / status
Low-friction entry and habit support · U1	Keep the short manual form and searchable history (FR03/FR05). Investigate reminders or a quick-add entry point as candidate enhancements, not approved MVP requirements.	Aisultan + Daryn; discovery / design pending.
Receipt trust, robustness and privacy · U2	Retain consent, review and manual fallback (FR07/FR08/FR14, NFR01). Measure labelled crumpled/local receipts and report failures before claiming that scanning is easier.	Dauren: evaluation; Daryn: recovery; Aisultan: notice/usability.
Useful but manually tracked savings · U3	Make the non-transfer/manual-update semantics explicit (FR12/FR13). Demonstrate a populated budget and goal. Investigate a cash-runway view separately; do not imply account connectivity.	Daryn + Aisultan; populated scenario evidence pending.
Duplicate bookkeeping · U4	Retain validated CSV import and account isolation (FR02/FR06). Validate the cash/receipt niche rather than promise bank parity. Bank sync remains outside the present scope.	Team scope decision proposed; no bank integration added.
Explainable, reliable analytics · U5	Evaluate real labelled descriptions and irregular histories (FR09/FR10, §9). Show that a suggestion is editable and a score uncalibrated. Investigate a truthful rationale display; do not invent explanations.	Dauren + Aisultan; quality/explanation design pending.

What changed in the project understanding

The feedback refines the initial “individual KZT user” into distinct groups. Receipt holders have a concrete capture/review need, while existing bank-app users may have little incentive to duplicate records. This narrows the proposed value proposition and makes privacy, correction effort and robustness central evaluation questions. Positive comments on controls or exact money remain opinions, not independent acceptance results.

Follow-up and evidence maintenance

Aisultan should retain the 24 October social-media survey record, associate each response with a non-identifying response ID, document consent and collection conditions, and preserve follow-up questions and conflicting answers. Link findings and decisions to requirements/issues, including the existing sessions task below. Do not backdate new feedback into the frozen tag or treat its old issue body as the latest complete evidence record.

https://github.com/KaberDaryn/smart-finance-project/issues/10

The later paired usability study (§9) remains a separate task: it must include actual task completion, waiting, correction and error measurements. Five qualitative survey responses alone do not establish speed gains, accuracy or task success rates.

3. Literature review

Review method and coverage

This is a targeted narrative review of personal informatics, human–AI interaction, document understanding, text classification and finance context. It contains **12 scientific publications, including 8 from 2022–2024**, within the five-year window before this submission. Four older papers are retained as explicit foundations. It is not claimed to be an exhaustive systematic review or a replicated model benchmark.

Critical synthesis

From finance context to usable workflows. Financial-literacy research motivates understandable information, not a claim that this software increases wealth [R1]. Personal informatics links collection with reflection/action [R9]. Consequently, the prototype joins entry with history, summaries and planning; five user-feedback summaries now provide reported relevance and adoption concerns (§2), while measured outcomes remain open.

Document pipelines are not interchangeable. Layout-aware methods [R2, R6], OCR-free end-to-end approaches [R3] and OCR recognition [R4] solve different stages. Pix2Struct [R5] and DocLLM [R7] broaden visual/layout approaches but have input/data assumptions unlike the current Gemini API. An OCR engine alone is not a complete merchant/date/total extractor. Alternatives must be compared as complete pipelines on the same labelled receipts.

Correct structure is not correct content. Hallucination research [R8] motivates reference checks rather than trusting valid JSON. Human–AI guidance [R10] motivates correction and recovery; SROIE [R12] motivates stage/field-specific evaluation. None proves that this prototype is accurate or faster than manual entry. The design therefore requires explicit review before save.

Transparent baselines need independent evaluation. Naive Bayes [R11] is a defensible inspectable baseline, not evidence of high quality. The tag's synthetic accuracy of 0.40 and macro-F1 of 0.4267 (§9) discourage stronger claims. Forecast performance needs chronological evaluation against a naive baseline [R13].

Additional methodological / engineering references

[R13] Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*, 3rd ed., §5.2. https://otexts.com/fpp3/simple-methods.html

[R14] Flutter. Architectural overview. https://docs.flutter.dev/resources/architectural-overview

[R15] SQLite. Appropriate uses for SQLite. https://www.sqlite.org/whentouse.html

[R16] OWASP. Password Storage Cheat Sheet. https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html

3.1 Recent scientific sources: context and extraction

R1 · Lusardi, A., & Mitchell, O. S. (2023). The Importance of Financial Literacy: Opening a New Field.

Journal of Economic Perspectives, 37(4), 137–154.

https://doi.org/10.1257/jep.37.4.137

Evidence / limitation	Project implication
Financial knowledge is relevant to economic decisions. A review of the field does not establish that this prototype improves literacy or wealth.	Motivates understandable summaries and planning; evaluate task outcomes, not an untested financial-benefit claim.

R2 · Huang, Y., et al. (2022). LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking.

ACM International Conference on Multimedia, 2022.

https://www.microsoft.com/en-us/research/publication/layoutlmv3-pre-training-for-document-ai-with-unified-text-and-image-masking

Evidence / limitation	Project implication
Combines text, image and layout pre-training. Published benchmark gains depend on their datasets and fine-tuning; they are not KZT receipt results.	Consider layout-aware baselines if resources permit. Current code uses an external provider, not LayoutLMv3; preserve a labelled local evaluation set.

R3 · Kim, G., et al. (2022). OCR-Free Document Understanding Transformer.

ECCV 2022, LNCS 13688, 498–517 (Donut).

https://doi.org/10.1007/978-3-031-19815-1_29

Evidence / limitation	Project implication
End-to-end document understanding avoids a separate OCR stage. It still requires model/data resources and validation on the relevant receipt layouts.	Treat image-to-structure as one alternative, not proof that OCR is unnecessary in every setting. Compare final fields and end-to-end latency.

R4 · Li, M., et al. (2023). TrOCR: Transformer-Based Optical Character Recognition with Pre-trained Models.

Proceedings of AAAI, 37(11).

https://doi.org/10.1609/aaai.v37i11.26538

Evidence / limitation	Project implication
Transformer text recognition operates at the recognition stage. Correct transcription alone does not identify which number is the purchase total.	Separate OCR/text quality from merchant/date/total extraction. A local OCR alternative also needs detection, parsing and field validation.

3.2 Recent scientific sources: layout and reliability

R5 · Lee, K., et al. (2023). Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding.

ICML, Proceedings of Machine Learning Research, 202, 18893–18912.

https://proceedings.mlr.press/v202/lee23g.html

Evidence / limitation	Project implication
Screenshot-to-structure pre-training supports visually situated understanding. Results on visual tasks are not a usability study of a finance application.	Use this as evidence of a visual-document approach, not evidence that this UI saves time. Measure human correction and navigation separately.

R6 · Tu, Y., et al. (2023). LayoutMask: Enhance Text-Layout Interaction in Multi-modal Pre-training for Document Understanding.

Proceedings of ACL 2023, 15200–15212.

https://doi.org/10.18653/v1/2023.acl-long.847

Evidence / limitation	Project implication
Text-layout interactions and position-aware masking target document understanding. External benchmarks do not resolve local-language/layout coverage.	Record receipt language, layout and reference-field coverage. Do not replace local field checks with a paper's aggregate benchmark score.

R7 · Wang, D., et al. (2024). DocLLM: A Layout-Aware Generative Language Model for Multimodal Document Understanding.

Proceedings of ACL 2024, 8529–8548.

https://aclanthology.org/2024.acl-long.463/

Evidence / limitation	Project implication
Combines text with spatial layout rather than an expensive image encoder. Its input assumptions differ from a raw-image provider API.	Compare complete pipelines including OCR/layout preparation, not model inference in isolation. DocLLM is not integrated in the submitted code.

R8 · Ji, Z., et al. (2023). Survey of Hallucination in Natural Language Generation.

ACM Computing Surveys, 55(12).

https://doi.org/10.1145/3571730

Evidence / limitation	Project implication
Reviews hallucination causes and evaluation across NLG. Its general taxonomy does not quantify hallucinations for the current receipt provider.	Valid JSON is not evidence of a correct date or total. Require reference labels, manual correction, explicit confirmation and failure recovery.

3.3 Foundational sources and design traceability

R9 · Li, I., Dey, A., & Forlizzi, J. (2010). A Stage-Based Model of Personal Informatics Systems.

Proceedings of CHI 2010, 557–566.

https://doi.org/10.1145/1753326.1753409

Evidence / limitation	Project implication
Distinguishes preparation, collection, integration, reflection and action. Foundational qualitative framing is not a causal test of this application.	Connect entry with review, history, summaries and planning. Observe breakdowns across the whole workflow rather than only extraction.

R10 · Amershi, S., et al. (2019). Guidelines for Human–AI Interaction.

Proceedings of CHI 2019, 1–13.

https://doi.org/10.1145/3290605.3300233

Evidence / limitation	Project implication
Human–AI design guidance emphasises understandable behaviour, correction and recovery. Applying guidelines is not the same as validated usability.	Show uncertain fields, allow edits and cancellation, and make manual fallback available. Test whether users can recover from missing/wrong proposals.

R11 · McCallum, A., & Nigam, K. (1998). A Comparison of Event Models for Naive Bayes Text Classification.

AAAI Workshop on Learning for Text Categorization.

https://aaai.org/papers/041-ws98-05-007

Evidence / limitation	Project implication
Explores event-model choices for text categorisation. A foundational baseline does not establish current multilingual finance classification quality.	Retain an inspectable multinomial NB baseline, fixed splits and macro-F1. Evaluate unknown tokens and class/language imbalance explicitly.

R12 · Huang, Z., et al. (2019). ICDAR2019 Competition on Scanned Receipt OCR and Information Extraction.

Proceedings of ICDAR 2019, 1516–1520 (SROIE).

https://doi.org/10.1109/ICDAR.2019.00244

Evidence / limitation	Project implication
Separates text localisation, OCR and key-information extraction. Benchmark task definitions do not make a small local convenience set representative.	Score merchant/date/total with field-specific reference coverage. Retain failed calls in primary denominators when reference labels exist.

4. Existing system comparison

Three relevant external systems are compared using eight consistent criteria. This is a desk review of official product/help pages, not a hands-on benchmark. **ND** means not established in the consulted sources, not “feature absent”. Platform, region and plan may change availability.

Criterion	YNAB [P1]	Wallet [P2–P3]	Money Manager [P4–P5]
1 · Capture	Bank import; transaction photos/memos	Manual cash records and bank sync	Manual income, expense and transfer
2 · Budgets	Spending/category planning	Category budgets and limit alerts	Budget versus actual spending
3 · Savings goals	Targets and progress	Financial goals mentioned; exact update semantics ND	Dedicated goal workflow ND
4 · Summaries	Spending / net-worth reporting	Cash-flow and spending insights	Category proportions / monthly patterns
5 · Navigation	Exact search/filter behaviour ND on consulted page	Exact search/filter behaviour ND on consulted pages	Search and category/account filtering
6 · Repeated-entry help	Import and attachments; structured receipt OCR ND	Automatic categorisation; exact receipt review flow ND	Bookmarks and recurring entries
7 · Future orientation	Loan planner / targets; comparable expense forecast ND	Cash-flow forecasting described	Recurring schedule; comparable forecast model ND
8 · Bank-independent path	Manual path limits ND in consulted page	Manual cash path documented	Manual records documented

Position of Smart Finance

The proposed prototype combines manual/CSV entry, editable receipt proposals, exact KZT storage and inspectable local analytics. It excludes bank integration and real transfers. Its budget/goal code is not evidence of a completed UI scenario. This narrower, testable integration scope is the project rationale—not a claim of feature novelty, competitor inferiority or proven market demand.

No price, Kazakhstan bank compatibility, structured receipt OCR quality or numerical product ranking is inferred. Missing evidence must not be converted into a competitor “No”.

4.1 Comparison evidence and design decisions

Official product source records

[P1] YNAB features
https://www.ynab.com/features

[P2] Wallet / BudgetBakers personal-finance solutions
https://budgetbakers.com/en/solutions/personal

[P3] Wallet budgets
https://budgetbakers.com/en/products/wallet/features/budgets/

[P4] Realbyte Money Manager: Home tab
https://help.realbyteapps.com/hc/en-us/articles/360046150874-Home-tab

[P5] Realbyte Money Manager: Stats tab
https://help.realbyteapps.com/hc/en-us/articles/360045310273-Stats-Tab

Interpretation

Observation	Design response
Recording, categories and budgets already exist.	Evaluate correctness and integration instead of presenting common features as inventions.
Automation can mean imports, templates or categorisation.	Keep the receipt proposal/review path optional and define exactly which input effort is being compared.
Published features do not measure local usability.	Use the same scripted tasks, named platform/build/plan and recorded timing if a later empirical product comparison is undertaken.
Requirements differ across financial products.	Do not calculate a weighted “winner” from ND values or compare this student prototype with full banking functionality.

Follow-up: verify KZT support and local bank compatibility only if those features enter scope; use the supplied qualitative feedback (§2) and further documented follow-up before claiming broad unmet demand. The present comparison supports a bounded engineering direction, not a commercial superiority claim.

5. Technology comparison

Four key decisions are considered, each with three reasonable candidates. Choices below are qualitative and driven by requirements, existing code and verification effort. No candidate performance benchmark or migration was executed.

T1 · Client implementation

Candidate	Fit / cost	Decision
Flutter / Dart	Existing client; shared UI toolkit [R14]. Camera, accessibility and native build behaviour still require target-specific checks.	Retain; validate each claimed target.
React Native / JS or TS	Reasonable cross-platform alternative; team could reuse JS expertise. Existing Dart widgets/validation would need rewriting.	Not selected for this iteration.
Native Android / Kotlin	Direct Android platform integration. Separate implementation for other targets; changes the cross-platform scope.	Reconsider if Android-only needs dominate.

The separate JavaScript client in web/ is present at the tag. It is not assumed to be a Flutter web build or feature-equivalent to the Dart client. Toolkit support is not evidence of a successful device run.

T2 · Server/API platform

Candidate	Fit / cost	Decision
Python ThreadingHTTPServer	Existing implementation, explicit routing and few framework dependencies. Team must own validation, errors, authentication and operational hardening.	Retain for bounded local demonstration; not a production-server recommendation.
FastAPI	Typed request models and generated API descriptions are useful for a growing schema. Requires migration and regression of money/auth semantics.	Reconsider for hosted API growth.
Django + REST framework	Structured ORM/admin/auth ecosystem. Larger restructuring and dependency burden than this prototype currently needs.	Reconsider for substantial account/admin workflows.

Criteria: implementation reuse, target coverage, platform integration, schema validation, migration risk, maintainability and verification effort. No security or speed superiority of the standard-library server is claimed.

5.1 Storage and receipt extraction alternatives

T3 · Relational financial storage

Candidate	Fit / cost	Decision
SQLite	Embedded, single-file, low setup effort [R15]. One writer at an instant; backup/write contention require attention.	Default local path.
PostgreSQL	Client/server option; adapter supplied. Requires service configuration, credentials, migrations and separate persistence/concurrency tests.	Optional path; not verified by SQLite checks.
MySQL	Viable relational client/server alternative. No supplied adapter; SQL/type/migration compatibility must be implemented and tested.	Not selected; no performance ordering claimed.

T4 · Receipt extraction pipeline

Candidate	Fit / cost	Decision
External Gemini adapter	Existing image → proposed JSON integration. Network, model changes, quota, latency and privacy constraints.	Optional assistance; server-side key, consent and manual fallback.
Tesseract + parsing	Can keep inference local. Needs preprocessing, OCR and reliable merchant/date/total extraction across layouts.	Alternative for offline/privacy needs; unbenchmarked.
PaddleOCR + field stage	Local document pipeline with replaceable extraction. Runtime/model packaging and labelled field evaluation still needed.	Revisit if local inference is a requirement.

All options must preserve integer minor-unit amounts, account boundaries, atomic saves and no-save-before-confirmation. OCR text recognition and field extraction are distinct tasks [R4, R12]. Review password work factors and key management, not merely the existence of hashing [R16].

Candidate documentation: https://reactnative.dev/docs/getting-started · https://developer.android.com/kotlin · https://fastapi.tiangolo.com/ · https://www.django-rest-framework.org/ · https://www.postgresql.org/docs/ · https://dev.mysql.com/doc/ · https://tesseract-ocr.github.io/ · https://www.paddleocr.ai/

6. Requirements specification

Proposed bounded MVP baseline. **M = Must** for the claimed local-demo scope; **S = Should** for proposed performance/accessibility targets. Mandatory acceptance is not inferred from code presence. Owners: D = Daryn, A = Aisultan, Da = Dauren. Vxx identifies a verification case in §9. Supervisor acceptance of this specification is pending.

Requirement / priority	Acceptance criterion	Check / owner
FR01 · M Accounts and sessions	Given a new valid username/password, registration returns 201; correct login returns a session; duplicate username is rejected; after logout that token cannot access private routes.	V01 D
FR02 · M Account isolation	Given users A and B, B cannot list A's transactions/goals/budgets or update A's goal; missing/invalid credentials return 401 on private routes.	V02 D
FR03 · M Record income and expenses	A valid dated income or expense returns 201 and appears in the same user's GET /transactions with matching kind, amount, category and description; invalid kind/amount/date is rejected without insertion.	V03 D
FR04 · M Exact money and summaries	Saving 1500.01 KZT produces 150001 integer minor units; after 2000.00 income and 1500.01 expense, income=200000, expense=150001 and balance=49999 minor units; retrieval/restart preserves values.	V04 D
FR05 · M History navigation	With two expenses and one income, All displays 3 records, Expenses 2 and Income 1; a unique description search returns only its matching entry; clearing search restores the selected filter's records.	V05 D + A
FR06 · M CSV validation and repeat detection	A valid two-row CSV imports both rows; any invalid row causes zero rows from that batch to be saved; importing identical CSV content again adds zero records and identifies the repeated import. Changed-but-similar files are not guaranteed duplicate detection.	V06 D
FR07 · M Consent and provisional receipt extraction	With consent unchecked, camera/gallery upload actions are disabled; an authenticated valid upload may return merchant/date/total/currency proposals but must not create a transaction. A rejected/unauthenticated upload must not invoke the provider.	V07 D + A

FR02 applies to all private data operations. FR04 includes a restart round-trip, not just immediate API retrieval. Consent is an application requirement; unauthorised upload rejection must occur before an external request.

6.1 Functional requirements: review, analytics and planning

Requirement / priority	Acceptance criterion	Check / owner
FR08 · M Receipt review and confirmed category reuse	For a proposal with missing date/category, Save is blocked with actionable messages; corrected valid fields are saved only after explicit confirmation. Saving a confirmed merchant/category makes it available for that user's next matching merchant; cancellation/rollback does not teach memory, and another user sees no mapping.	V08 D
FR09 · M Description category suggestion	A known English description returns an editable category suggestion and identifies the score as uncalibrated; all-unknown tokens return no category; accepting or changing the suggestion does not save a transaction until Save is pressed. Real-world quality is evaluated separately.	V09 Da + D
FR10 · M Expense estimate	For completed-month expense totals 100, 200 and 300 KZT the next estimate is 200 KZT; current-month expense/income is excluded. Missing months after first expense count as zero; insufficient-history state is explicit.	V10 Da
FR11 · M Category and monthly insights	For 5478.00 and 680.00 KZT expenses in Groceries, displayed expense total is 6158.00 and category share is 100%; income must not increase expense shares. Empty history displays an understandable empty state.	V11 Da + D
FR12 · M Monthly budgets	After saving a 1000.00 KZT overall/category budget, retrieval shows that limit for that user/month. A 1200.00 matching expense gives spent=1200.00 and an exceeded state; unrelated category/month records do not consume a category-specific limit.	V12 D + Da
FR13 · M Savings goals	Create target 10000.00 KZT and set saved amount to 2500.00; retrieval/UI show the updated amount and 25% progress. The operation does not insert a financial transaction or change ledger balance; another user cannot update the goal.	V13 D
FR14 · M Provider failure recovery	Mock missing configuration/quota, timeout and malformed response; API returns the documented controlled error (503/504/502 respectively), server remains usable, UI explains recovery and offers manual entry. No expense/category memory is saved by a failed scan.	V14 D + A

A category suggestion is not financial advice; merchant/category reuse is a separate per-user memory mechanism, not online classifier training. Forecast and classifier real-world quality are separate evaluation questions (§9).

6.2 Non-functional requirements and change control

Requirement / priority	Acceptance criterion	Check / owner
NFR01 · M Secrets and credential handling	Repository/client bundles/logs contain no real provider keys, plain-text passwords or usable raw session tokens. Inspect salted password hashing/work factor and hashed stored sessions; invalid/expired/revoked tokens are rejected. Any discovered secret must be removed from distribution and rotated.	V15 D
NFR02 · M Input and storage integrity	Reject receipt input over 5 MiB and invalid content types before provider invocation; enforce integer-money/date bounds and atomic transaction+merchant-memory writes. Failing an insertion leaves neither half committed.	V16 D
NFR03 · S Responsiveness and availability measurement	Proposed local acceptance target: for 30 sequential /health requests and authenticated summary reads on a seeded 1000-record account, each route has p95 <= 2 s and zero unhandled errors. Record hardware/database/build and all timings. Receipt calls are measured separately.	V17 D + A
NFR04 · S Usability and accessible recovery	At 360 px viewport and 200% browser zoom, required form fields/error text and Save/manual-entry controls remain reachable without horizontal clipping; keyboard focus reaches applicable web controls. User tasks record completion/errors/correction effort rather than assuming marketing text demonstrates usability.	V18 A
NFR05 · M Reproducibility and deployment evidence	Each reported run records source commit, client build, runtime/database versions, commands, environment without secrets, and dated output. Verify SQLite persistence after restart and separately verify PostgreSQL before claiming that deployment path passes.	V19 D

Interpretation and release gates

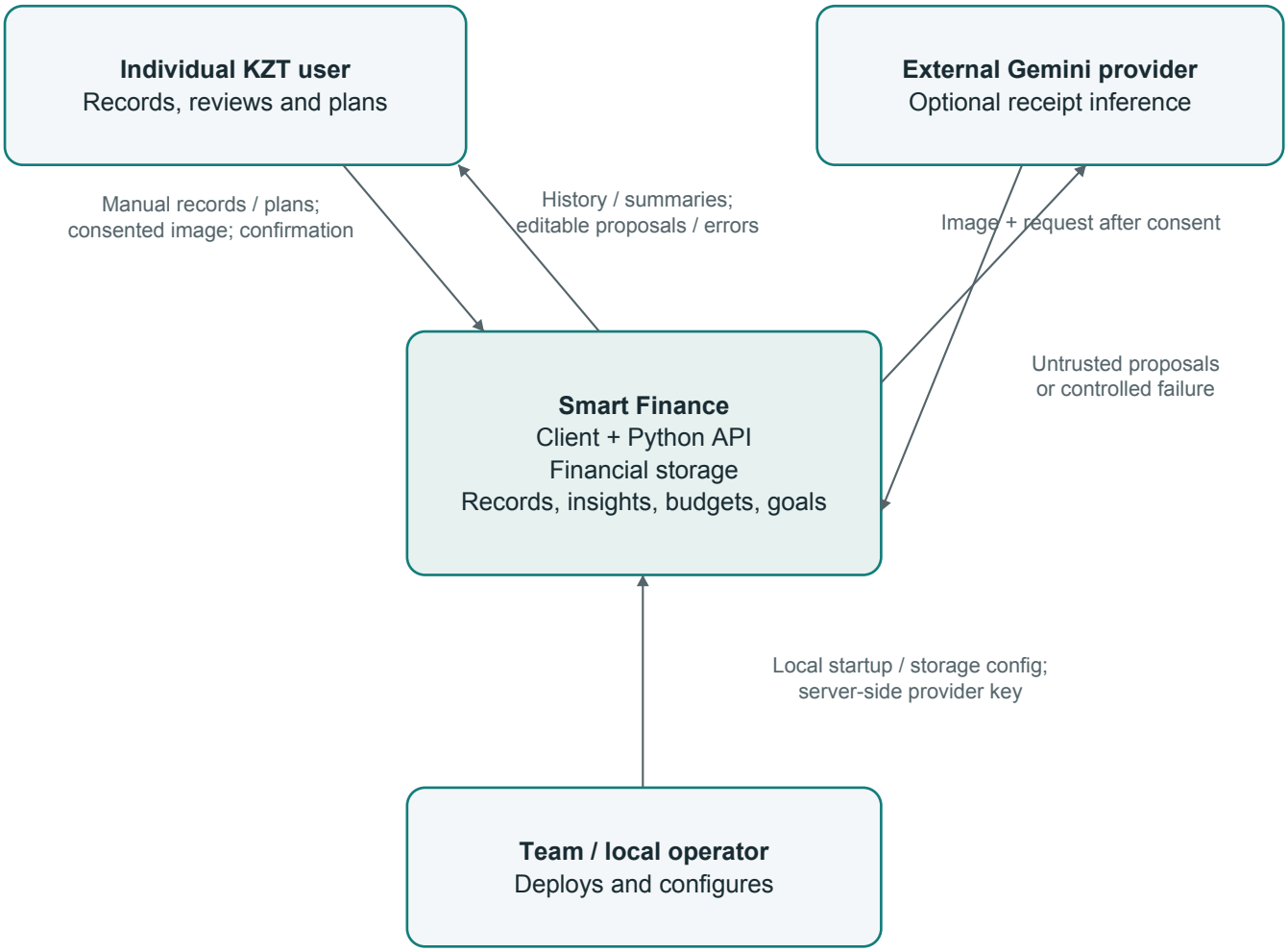
M requirements define local functionality/security/integrity expectations; they are not all declared passed. S targets require team/supervisor agreement before acceptance testing and cannot be silently relaxed after a failed run. The p95 target is a proposed local API target, not a measured result and not a receipt-provider latency target.

Track a failed criterion as a defect or an explicit scope decision. Retain original evidence, record the revised requirement, reason and reviewer, then repeat affected tests. Native-device and PostgreSQL claims require separate platform/deployment evidence even when local SQLite tests pass.

7. Architecture and diagrams

The context, use-case and component views are consistent with the local prototype boundaries. Diagrams are design documentation, not proof that every scenario passed. Editable sources are embedded in the PDF and support package; public source-file URLs are not available at the inspected tag.

7.1 System Context



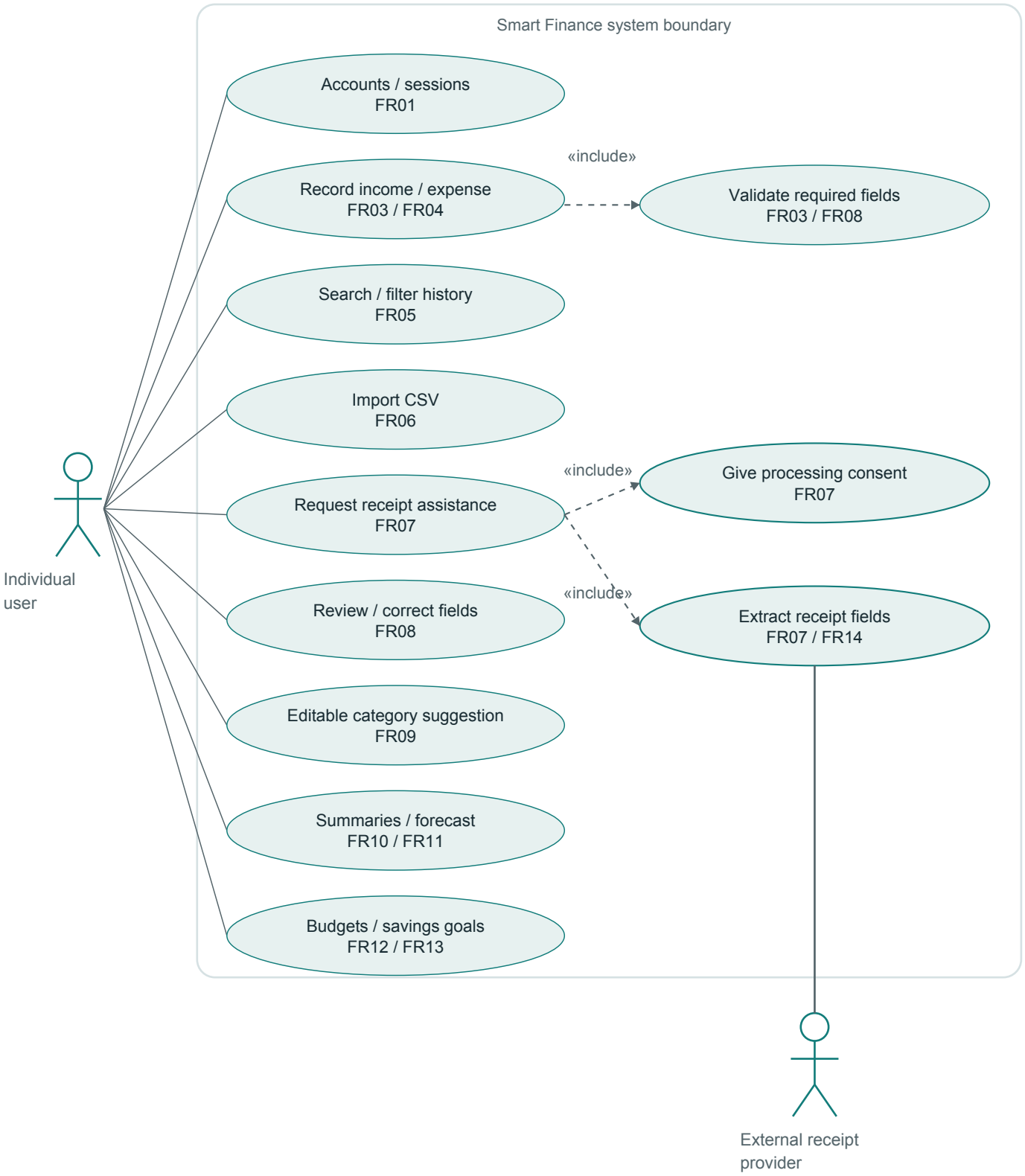
Trust boundaries: user input and provider output are untrusted. The server authenticates access and validates fields; the user confirms the transaction. The provider key stays server-side. Core manual entry must remain usable without receipt assistance.

Data boundary: the ledger stores KZT minor units and user-scoped records. A displayed balance is income minus recorded expenses, not a connected bank balance. Receipt upload sends data to an external service; consent and minimisation are required.

Editable attachment: system-context.mmd. Public code context:
https://github.com/KaberDaryn/smart-finance-project/blob/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/backend/server.py

7.2 Use Case Diagram (draft UML view)

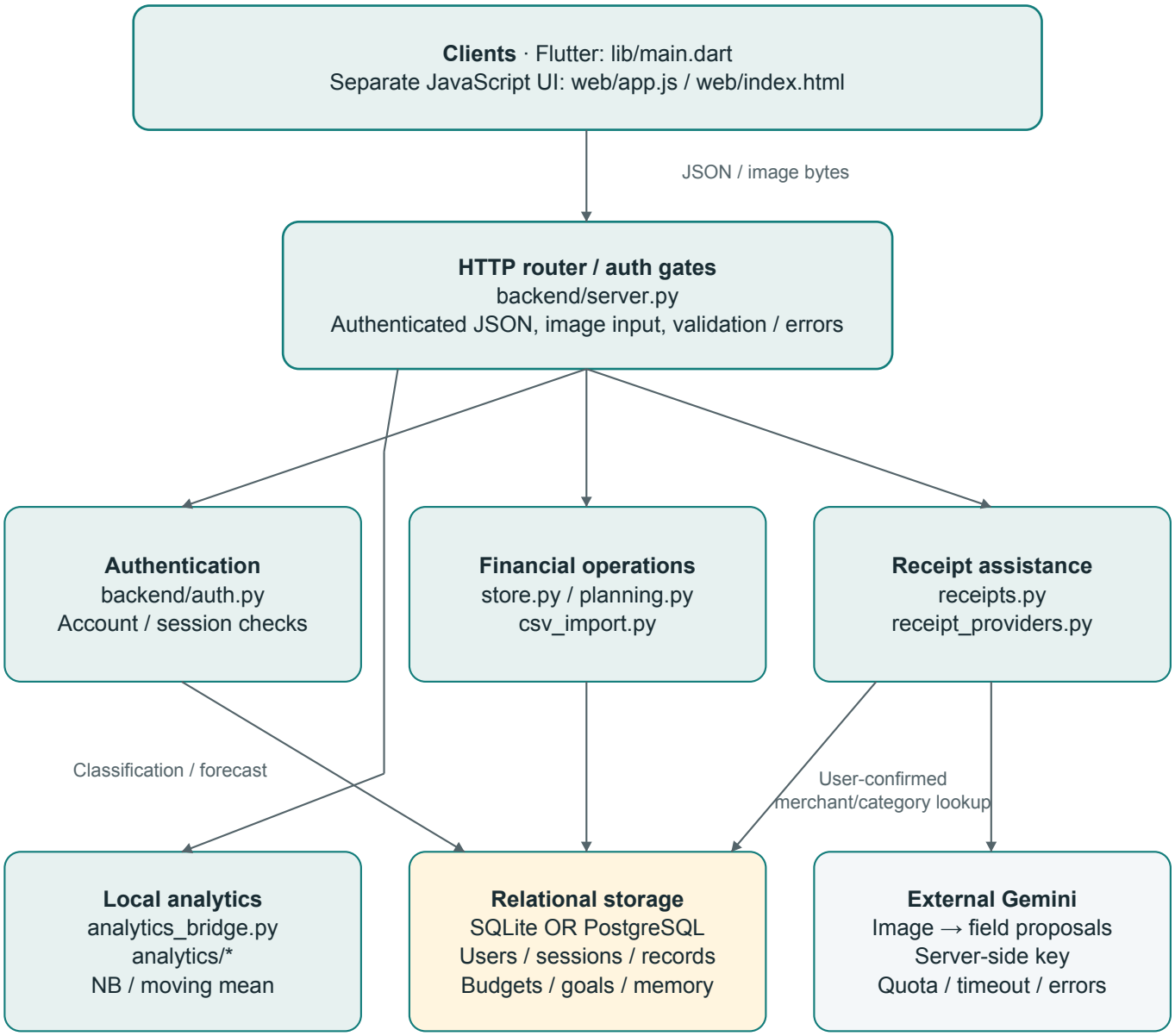
Individual user and external provider are actors; the rectangle is the Smart Finance boundary. Solid lines are actor associations; dashed arrows marked «include» denote mandatory included behaviour. Review is not an automatic save. FR02 is cross-cutting account isolation.



Editable attachment: use-case.puml. FR08 correction precedes a separate explicit save; this is not an «include» relation that would force every review to commit data.

7.3 Component Diagram (draft)

Modules and dependencies shown below describe the frozen source. SQLite is the verified local path; PostgreSQL is an alternative adapter, not a concurrently required second store. Client feature parity is not assumed.



Consistency: routes enforce authentication; storage and planning use account-scoped queries. Confirmed expense and merchant/category memory are committed atomically. Extraction alone does not insert a transaction. Analytics do not execute payments or savings transfers.

Money/data: records include id, kind, amount_minor, category, description, created_at and user_id. For example, 125000 minor units = KZT 1250.00. Source hashing is not a completed security audit; review work factors and token handling [R16].

Editable attachment: component.mmd. Source:
https://github.com/KaberDaryn/smart-finance-project/tree/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/backend/
API / storage docs:
https://github.com/KaberDaryn/smart-finance-project/blob/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/docs/API.md ·
https://github.com/KaberDaryn/smart-finance-project/blob/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/docs/DATABASE.md

8. UX / prototype

A reproducible local prototype is available in the frozen repository. No verified publicly hosted live URL or approved Figma artifact was supplied. The source/ZIP and setup links below provide access to the available implementation, not a claim of zero-setup hosting.

https://github.com/KaberDaryn/smart-finance-project/tree/prob-period-1/web
https://github.com/KaberDaryn/smart-finance-project/archive/refs/tags/prob-period-1.zip
https://github.com/KaberDaryn/smart-finance-project/blob/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/docs/RUNNING.md

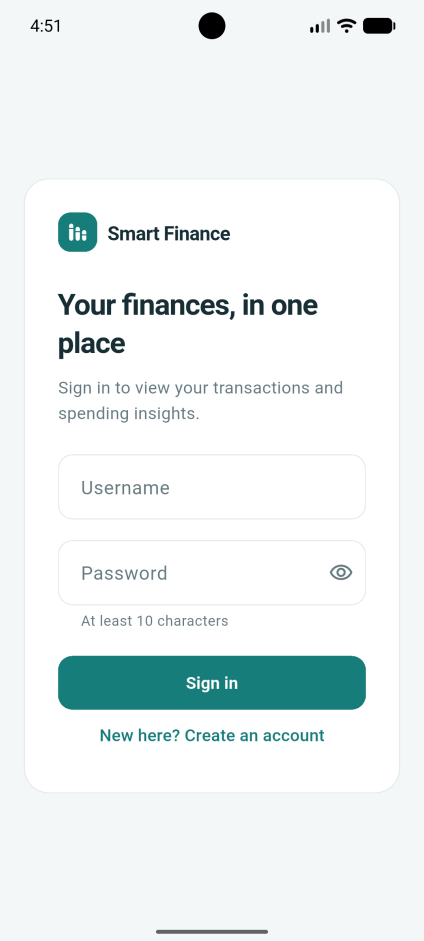


Figure 8a. Supplied sign-in state; fields contain no credentials.

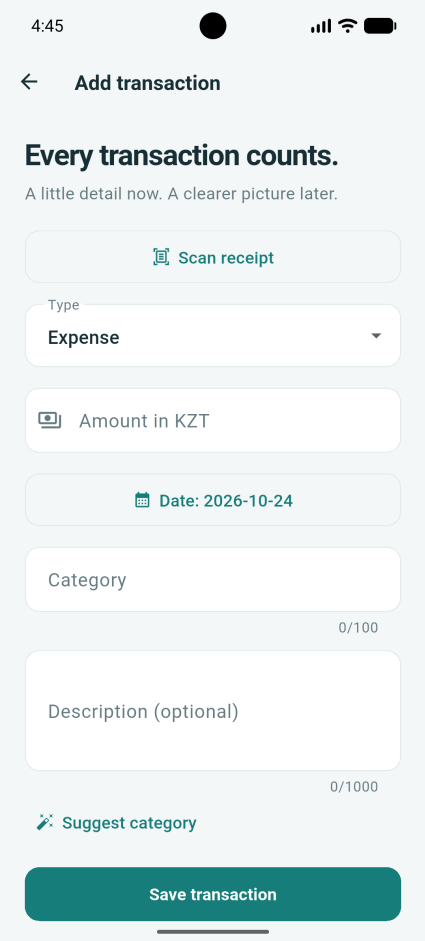


Figure 8b. Supplied manual-entry state; use it as a scenario illustration, not a completed acceptance test.

Scenario A: create/sign in to an account, enter a dated income or expense, review required fields, save, then retrieve it in the same account. A valid transaction should round-trip without losing decimals; another account must not see it (FR01–FR05).

The team confirms that the supplied report screens were captured during emulator-based interface checks. They document demonstrated UI states and flows. Emulator version, exact client build/commit, test steps and outcomes should be added to the evidence record; physical-device testing and performance measurements are not inferred.

8.1 Receipt capture, consent and review

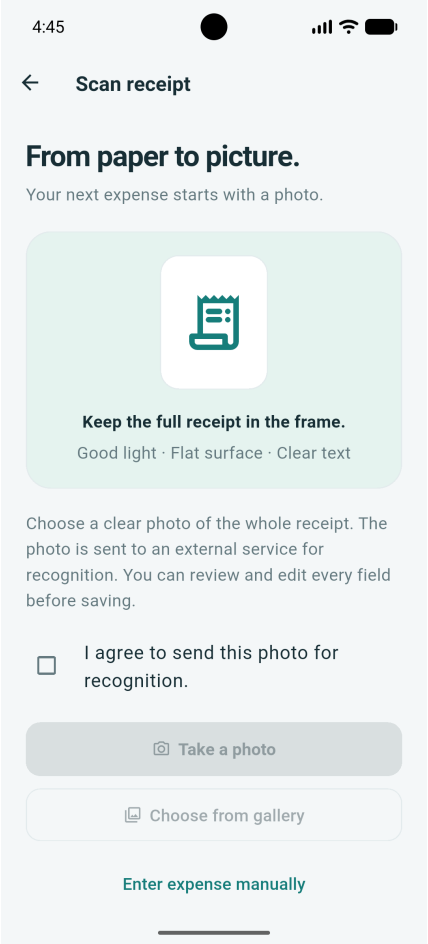


Figure 8c. Consent unchecked; capture/gallery controls disabled.

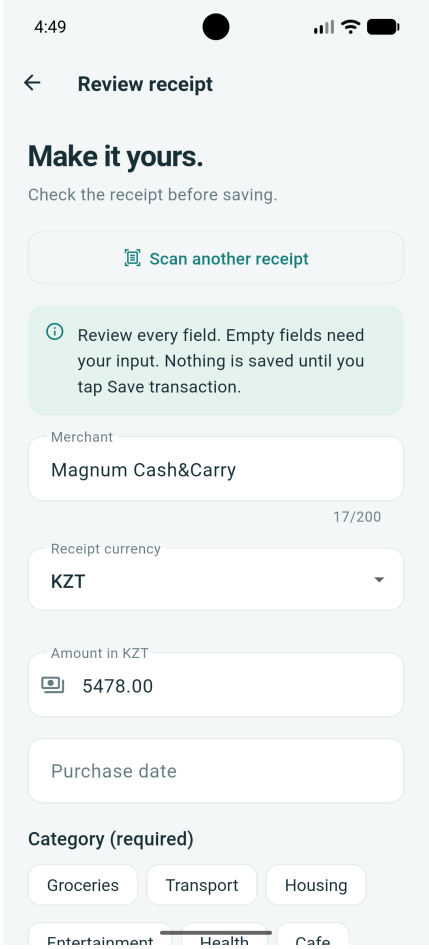


Figure 8d. Review shows Magnum Cash&Carry, KZT 5478.00 and an empty purchase date.

Scenario B · optional assisted entry

Explain external processing before enabling upload. After consent, capture/select an image and obtain provisional fields. Display merchant, currency, total and purchase date for correction; require category/date as appropriate. Nothing should enter the ledger until explicit Save. A provider failure must leave manual entry available (FR07/FR08/FR14).

A displayed merchant or amount does not establish the correctness of every extracted field or the provider's response time. A later manually completed date is not ground truth for the original printed receipt. Do not infer a live model-quality result from UI screenshots.

> Version limitation: the tag's separate JavaScript currency-change handler uses Math.round(total), which can round away KZT decimals. This is a specific release gate for the 1234.56 → 123456 minor-unit scenario, not a claim that every client path has the same defect. See issue #21 and §9.

https://github.com/KaberDaryn/smart-finance-project/blob/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/web/app.js
https://github.com/KaberDaryn/smart-finance-project/issues/21

8.2 Validation, saved records and spending reflection

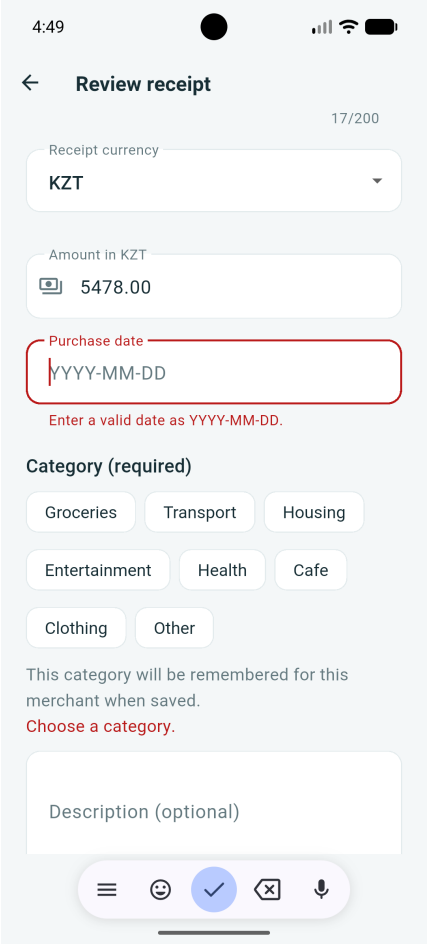


Figure 8e. Required purchase-date/category errors are visible.

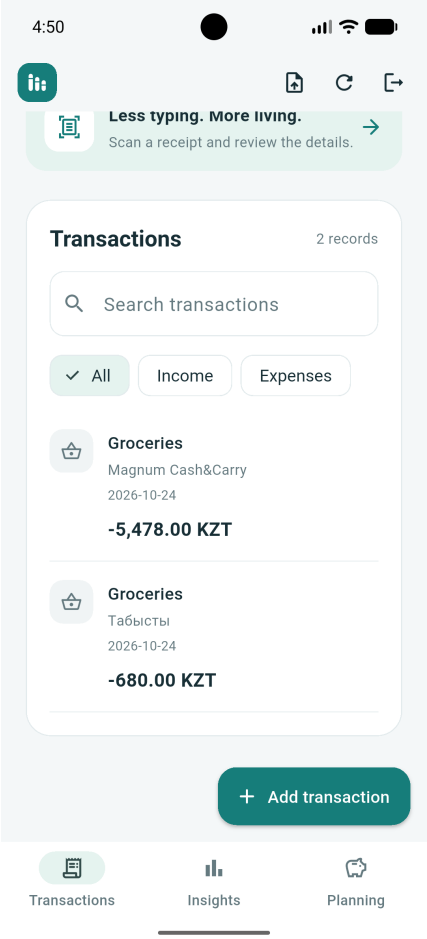


Figure 8f. Later list state shows KZT 5478.00 and KZT 680.00 expenses.

Scenario C · validate, confirm and inspect

With date/category missing, Save should be blocked with actionable errors. After correction and confirmation, retrieve the record and check exact amounts and account ownership. The two visible expenses sum to KZT 6158.00. This arithmetic check does not prove an uninterrupted save action or validate all charts.

User-confirmed merchant/category reuse must remain user-scoped. Cancellation or failed storage must not update memory. Category suggestions can be edited; the stored category is the user's confirmed decision, not an autonomous financial action.

Planning scenario still to demonstrate

Create a monthly budget, add spending beyond its limit, inspect the exceeded state, then create/update a savings goal. Updating saved progress must not change ledger balance. The supplied empty Planning state alone is insufficient evidence for these completed scenarios (FR12/FR13).

9. Verification and evaluation plan

9.1 Reproduced checks on the exact tag

Run record	Observed result
Source / environment	8dd1d51679fe4672bb00e198b3a55ccea2d0c774 Linux · Python 3.13.15 · temporary SQLite / local HTTP tests
Run time	24 October 2026, 11:42 UTC (verification manifest)
Automated command	python -m unittest discover -s tests -v
Test result	**50 tests in 10.443 seconds — OK**
Receipt provider	Not called; mocked checks do not measure live extraction quality.
Analytics command	python -m analytics.evaluate

The local run used the tag's backend, analytics and test files. It was not a Flutter build, a public load test, a PostgreSQL deployment or an independently executed emulator/device or participant study. The supplied screenshot reporting 51 tests is not attributed to this frozen build and is not substituted for the reproduced 50-test result.

9.2 Synthetic analytical fixtures: reproduced, not research findings

Fixture	Result / limitation
Classification	40 synthetic English descriptions; 30 train / 10 test; seed 42; five classes. Accuracy 0.4000; macro-F1 0.4267. Small synthetic data are not evidence of target-user/language quality.
Forecasting	12 synthetic monthly totals; four expanding-window holdouts. Three-month-mean MAE = KZT 1250; last-month baseline MAE = KZT 1375. Four synthetic outcomes do not establish a reliable real-world improvement.

Supporting attachments: tag-tests.log, tag-analytics.json, verification-run.json. Public test/evaluation sources: https://github.com/KaberDaryn/smart-finance-project/tree/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/tests/ · https://github.com/KaberDaryn/smart-finance-project/blob/8dd1d51679fe4672bb00e198b3a55ccea2d0c774/analytics/evaluate.py

9.3 Acceptance execution and traceability

These Vxx cases specify release acceptance. The automated run supports backend behaviour but is not a blanket PASS for full UI, native, performance or restart criteria. Owners are defined in §6; preserve raw logs and build IDs for every run.

Case	Area	Execution / measurement	Requirement
V01–V02	Accounts and isolation	Register/login/logout; two accounts; reject absent/revoked token; no cross-user read/update.	FR01–02
V03–V04	Records and exact money	Valid/invalid kind/date/amount; 1500.01 round-trip; income 2000 → balance 499.99; repeat after restart.	FR03–04
V05–V06	History and CSV	Search/filter against known fixture; valid import; invalid batch leaves no partial writes; defined duplicate handling.	FR05–06
V07–V08	Receipt and memory	No provider call before auth/consent; no save by scan; correct missing fields; explicit save; scoped memory; rollback/cancel teach nothing.	FR07–08
V09–V11	Analytics and summaries	Editable known/unknown-token category; completed-month mean; 5478+680=6158; income excluded from expense shares.	FR09–11
V12–V13	Planning	Budget 1000 / spent 1200 → exceeded; goal 10000 / saved 2500 → 25%; no balance change.	FR12–13
V14	Provider recovery	Mock configuration/quota, timeout and malformed output; controlled 503/504/502; manual fallback; no partial record.	FR14
V15–V16	Security / integrity	Secrets and work-factor review; token invalidation; invalid MIME / >5 MiB blocked before provider; atomic save.	NFR01–02
V17–V18	Performance / accessibility	30 local health + summary samples on 1000 records; p95; 360 px / 200% zoom / keyboard recovery. Targets require approval.	NFR03–04
V19	Reproducibility	Commit/client/runtime/DB evidence; SQLite restart; separate PostgreSQL run before that deployment is claimed.	NFR05

Specific browser release gate: issue #21’s decimal/currency and receipt-integration scenarios must be checked in the named JavaScript build, independently of Flutter CI or backend unit-test success.

9.4 Outcome metrics, conditions and baselines

Question	Protocol, metric and baseline
Receipt field quality	Use a consented fixed labelled receipt set covering relevant language/layout conditions. Independently review merchant/date/total labels and mark unknown fields. Freeze model identifier, prompt, timeout/retries, image preparation and build. Exact/normalised match per labelled field; failed calls count as unmatched where labels exist. Report sample count, coverage, service failures and uncertainty.
Receipt latency	Retain all attempt times and success/failure status; report mean, median and nearest-rank p95. Include preprocessing/network/provider stages as defined. These are pipeline times, not human task times. No unmeasured latency target is declared passed.
Classifier quality	Independently label real descriptions, fix train/test split before tuning, avoid duplicates/leakage and report class/language coverage. Accuracy + macro-F1 + confusion matrix + unknown-token rate. Compare NB with majority-class and a simple alternative classifier on the same held-out set.
Forecast quality	Use chronological expense histories and past-only windows. Document missing months as zero after first expense; exclude current incomplete month and income. Report MAE (KZT); MAPE only with explicit zero-actual exclusion/counts. Compare three-month mean against last-month naive baseline [R13].
Human effort / usability	Aisultan records real participants and consent; compare paired manual and receipt-assisted entry with counterbalanced task order. Include capture/upload/wait/review/correction/save in total time. Report completion, critical errors, corrections and subjective feedback; do not generalise from an undeclared small sample.

Decision rules and evidence handling

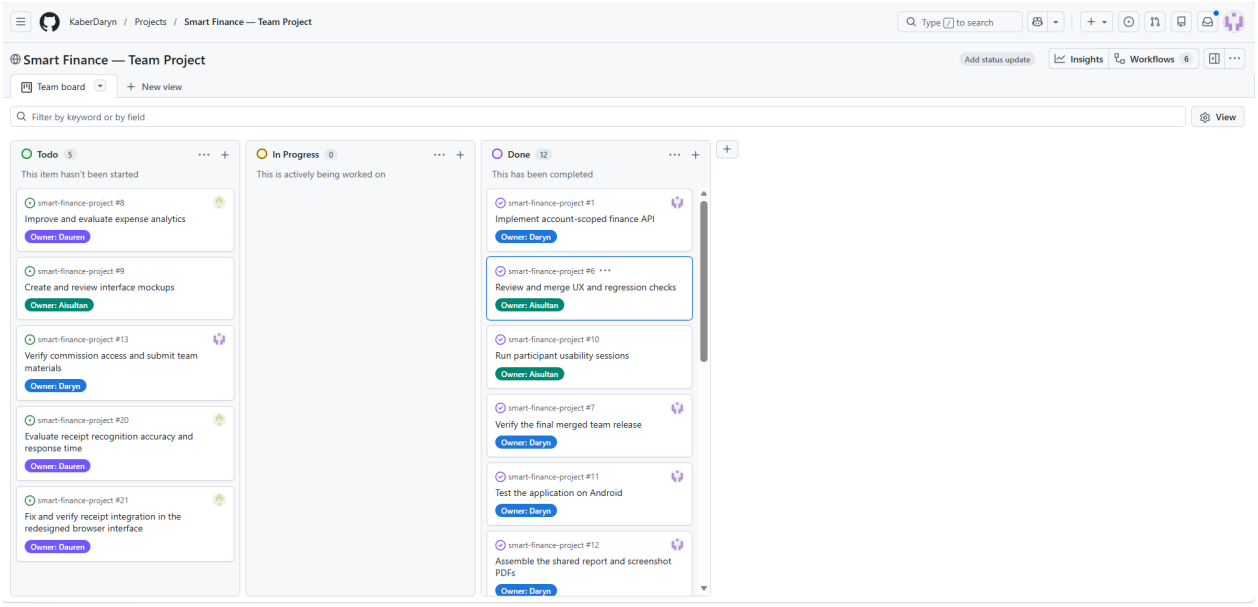
Mandatory functional criteria are pass/fail. For comparative quality, report observed differences with sample size and uncertainty; the team/supervisor must agree any quality threshold before a new held-out run. No threshold is retrofitted to the existing synthetic results. Use anonymised/redacted material, preserve raw outputs privately and publish only consented non-identifying summaries.

The previously supplied 16-receipt private pilot is not a verified run of this tag: run timestamp/build provenance and raw outputs are not established here. It is not merged with synthetic data or new UI screenshots to create an accuracy claim. Live extraction quality and comparative usability remain open.

10. Project management and risks

Repository: https://github.com/KaberDaryn/smart-finance-project
Frozen submission: https://github.com/KaberDaryn/smart-finance-project/tree/prob-period-1
Team board: https://github.com/users/KaberDaryn/projects/1

The supplied GitHub Projects snapshot shows **5 Todo, 0 In Progress and 12 Done**. This is a management snapshot, not a frozen tag test report. The board URL was supplied by the team; verify committee access separately. The prescribed YouTrack project was not supplied and approval to use GitHub Projects instead is not established.



Team member	Declared responsibility / evidence to retain
Kaber Daryn	Architecture, client, API, storage, integration, local deployment; trace to commits/PRs, tests and named client builds.
Nuriman Aisultan	UX/UI, QA/test planning, relevance discussions and usability materials; retain actual notes, review and observation results.
Akhmetov Dauren	Categorisation, forecast and analytical evaluation; retain datasets, split manifests, metrics and receipt-quality evidence.

Development and review traceability

The inspected history reachable from the tag contains 18 commits dated 19–24 October 2026 and author names Daryn Kaber, Daryn, aisusjan and daakhmetov. This is a short traceable history, not proof of long-term progression or line-by-line authorship. Merge commits reference PRs #14–#17. The queried PR #17 review list was empty; merge status alone does not establish teammate approval.

https://github.com/KaberDaryn/smart-finance-project/commits/prob-period-1 · https://github.com/KaberDaryn/smart-finance-project/pull/17

10.1 Task evidence, milestones and risk controls

Task record	Evidence interpretation / next action
#1 · API · Daryn	Closed; checked implementation criteria and commit link. Retain exact-tag tests and verify remaining UI acceptance.
#8 / #20 · Analytics · Dauren	Open. Synthetic metrics exist; independent real-data classifier/receipt quality still required.
#9 · Mockups · Aisultan	Open. Code-backed interface description is not approved mockup evidence.
#10 · Sessions · Aisultan	Old issue body says only protocol/blank observations. Five social-media survey responses dated 24 October 2026 are now supplied (§2); link the new survey record. Timed participant results remain unrecorded.
#11 · Android · Daryn	Task body records an earlier device-work deferral. Emulator checks are confirmed by the team and illustrated in the report screens; record build and steps. Physical-device testing is not claimed.
#21 · Browser receipts · Dauren	Open. Complete exact-decimal, missing-field, category-memory and failure-recovery browser checks.

Full task links: https://github.com/KaberDaryn/smart-finance-project/issues/1 · https://github.com/KaberDaryn/smart-finance-project/issues/8 · https://github.com/KaberDaryn/smart-finance-project/issues/9 · https://github.com/KaberDaryn/smart-finance-project/issues/10 · https://github.com/KaberDaryn/smart-finance-project/issues/11 · https://github.com/KaberDaryn/smart-finance-project/issues/20 · https://github.com/KaberDaryn/smart-finance-project/issues/21

Milestone / owner	Exit evidence (proposed dependency order)
M1 · Team + supervisor	Review the five supplied user summaries and teacher comments (§2/§11); confirm scope, decisions and acceptance targets; retain named/date-stamped review.
M2 · Daryn / all	Commit academic docs and editable diagrams; reconcile report/repo version; retain source/build logs without silently moving the submitted tag.
M3 · Daryn + Aisultan	Complete exact-money/restart and budget/goal acceptance; retain emulator build/steps/screens and access checks; reconcile task records with newer evidence.
M4 · Dauren + Aisultan	Complete independent quality and paired usability runs under §9; publish anonymised results.

Risk controls: provider quota/latency → manual fallback (Daryn); sensitive images → informed consent/minimisation (Aisultan); synthetic/bias risk → independent labels/held-out data (Dauren); client divergence → build-specific UI regression (Daryn); missing approvals/YouTrack/access → explicit supervisor decision and committee access check (team). Calendar deadlines remain to be agreed; no historical dates are invented.

11. Supervisor feedback

Source [F1]: teacher feedback received from the team on 24 October 2026, following review of the repository and report. The report's named supervisor is Bakayeva Bakyt, Senior Lecturer; the supplied feedback text itself does not identify its author or exact reviewed build. The text below is reproduced verbatim; it is not presented as a signed approval of every acceptance criterion.

11.1 Review text supplied by the team

"I looked through the repository and the report. The project is solid: the app actually runs, it has tests, Docker support and clear setup instructions, so everything can be tried out in practice. I liked that a scanned receipt never goes into the records automatically and is first shown for review, and that amounts are calculated exactly, with no lost decimals. The interface is clean and easy to follow, and the "photograph the receipt, correct it, save" scenario feels convenient. The report is tidy and clearly separates what has been verified from what is still planned.

There are a few things that could make the project even stronger. A couple of short conversations with real users would help show that the app solves a genuine problem. The receipt recognition, which is the most interesting part of the project, would benefit from a test on a small labelled set of receipts, with the accuracy reported in numbers. The category suggestions currently rely on a small test dataset and should become more accurate once they are trained on real expenses. It would also be handy to keep the diagrams and documents inside the repository so they can be opened by link, and to add a screenshot of the Planning screen with a budget and a savings goal.

Overall, this is a good foundation for a diploma project. With user feedback and measured recognition accuracy added, the work will look strong and complete."

Interpretation and review boundary

The review recognises runnable software, test/configuration support, a clear interface and user control over receipt saving. These are qualitative review judgements. Docker configuration is not the same as an independently verified deployment; praise for exact amounts does not override the tag-specific JavaScript currency-handler issue recorded in §8.1. The feedback does not supply the missing model-quality numbers or Planning scenario evidence.

Resulting revisions: five supplied user-feedback summaries and requirement implications have been added to §2; the full teacher review is retained here; actions, owners, acceptance evidence and remaining work are recorded on the following page. The user summaries are not retroactively attributed to the reviewed tag.

11.2 Responses to feedback and supporting materials

Teacher comment [F1]	Response / acceptance evidence	Owner / status
Hold short conversations with real users.	Five social-media survey responses from 24 October 2026 added, including negative findings and scope decisions (§2). Preserve consent and non-identifying response traceability; no timed study is claimed.	Aisultan; date/method and findings recorded; evidence retention open.
Measure recognition on a labelled receipt set.	Freeze build/model/prompt; use independently reviewed labels; report merchant/date/total matches, failures and latency under §9. The earlier private pilot is not relabelled as an exact-tag run.	Dauren + Daryn; matched-build evaluation pending.
Improve category suggestions with real expenses.	Collect consented labelled descriptions, freeze held-out split and compare accuracy/macro-F1 with baselines. Improvement must be measured, not assumed from adding data.	Dauren; dataset/evaluation pending.
Keep documents and diagrams in the repository.	Readable diagrams and editable sources prepared and embedded. Commit academic docs/exports/sources, update README and verify public links against the reviewed submission version.	Daryn / all; materials prepared, publication pending.
Show Planning with a budget and savings goal.	Run FR12/FR13: populated budget, overspending state, saved goal/progress and unchanged ledger balance; retain screenshots and build ID.	Daryn + Aisultan; scenario/screenshot pending.

Review closure

Retain the feedback received on 24 October 2026, confirm its author and reviewed version, link each action to its task and resulting commit, then request confirmation that the substantive comments were addressed. No completed publication, quality improvement, Planning scenario or formal sign-off is inferred from this response plan. GitHub Projects substitution for YouTrack still needs an explicit decision.

Embedded supporting materials

Open/save attachments in a compatible PDF reader. They include the original supplied feedback text, academic drafts, editable sources, diagram exports, board image and verification evidence. Attachments are accessible files, not public repository URLs.

1. support-package.zip — academic materials, USER_FEEDBACK.md, TEACHER_FEEDBACK.md and evidence
2. system-context.mmd — editable context source
3. use-case.puml — editable UML source
4. component.mmd — editable component source

Current status: local tests are traceable to prob-period-1; emulator checks, user feedback and the teacher review are recorded. Remaining work includes survey evidence retention and review closure, measured recognition/classification/usability, populated Planning evidence and reviewed repository publication.